

Mathematics Extension 2

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

**Total marks:
100**

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–18)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

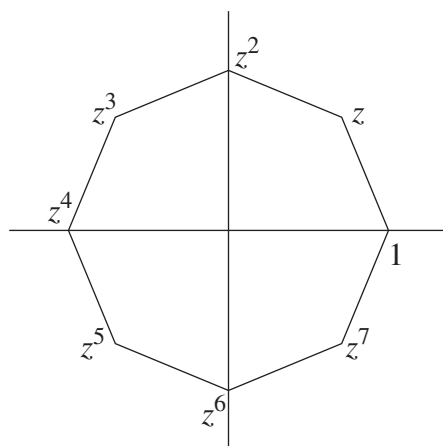
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 The complex number z is chosen so that $1, z, \dots, z^7$ form the vertices of the regular polygon shown.



Which polynomial equation has all of these complex numbers as roots?

- A. $x^7 - 1 = 0$
 - B. $x^7 + 1 = 0$
 - C. $x^8 - 1 = 0$
 - D. $x^8 + 1 = 0$
- 2 Suppose ℓ is a line and S is a point NOT on ℓ .

The point P moves so that the distance from P to S is half the perpendicular distance from P to ℓ .

Which conic best describes the locus of P ?

- A. A circle
- B. An ellipse
- C. A parabola
- D. A hyperbola

3 Which complex number lies in the region $2 < |z - 1| < 3$?

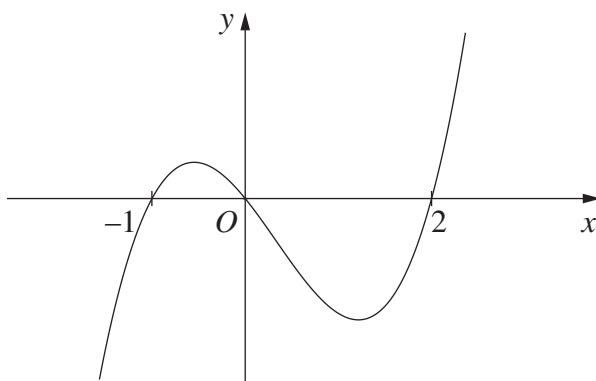
A. $1 + \sqrt{3}i$

B. $1 + 3i$

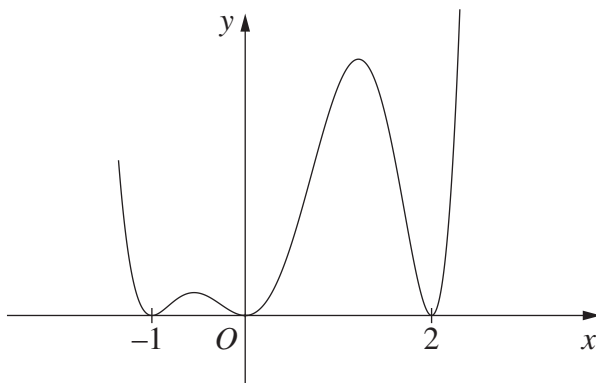
C. $2 + i$

D. $3 - i$

4 The graph of the function $y = f(x)$ is shown.



A second graph is obtained from the function $y = f(x)$.



Which equation best represents the second graph?

A. $y = \sqrt{f(x)}$

B. $y^2 = f(x)$

C. $y = [f(x)]^2$

D. $y = f(x^2)$

- 5 The polynomial $p(x) = x^3 - 2x + 2$ has roots α , β and γ .

What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

- A. -10
- B. -6
- C. -2
- D. 0

- 6 It is given that $z = 2 + i$ is a root of $z^3 + az^2 - 7z + 15 = 0$, where a is a real number.

What is the value of a ?

- A. -1
- B. 1
- C. 7
- D. -7

- 7 It is given that $f(x)$ is a non-zero even function and $g(x)$ is a non-zero odd function.

Which expression is equal to $\int_{-a}^a f(x) + g(x) dx$?

- A. $2 \int_0^a f(x) dx$
- B. $2 \int_0^a g(x) dx$
- C. $\int_{-a}^a g(x) dx$
- D. $2 \int_0^a f(x) + g(x) dx$

- 8 Suppose that $f(x)$ is a non-zero odd function.

Which of the functions below is also odd?

- A. $f(x^2)\cos x$
- B. $f(f(x))$
- C. $f(x^3)\sin x$
- D. $f(x^2) - f(x)$

- 9 A particle is travelling on the circle with equation $x^2 + y^2 = 16$.

It is given that $\frac{dx}{dt} = y$.

Which statement about the motion of the particle is true?

- A. $\frac{dy}{dt} = x$ and the particle travels clockwise
- B. $\frac{dy}{dt} = x$ and the particle travels anticlockwise
- C. $\frac{dy}{dt} = -x$ and the particle travels clockwise
- D. $\frac{dy}{dt} = -x$ and the particle travels anticlockwise

- 10 Suppose $f(x)$ is a differentiable function such that

$$\frac{f(a)+f(b)}{2} \geq f\left(\frac{a+b}{2}\right), \text{ for all } a \text{ and } b.$$

Which statement is always true?

A. $\int_0^1 f(x) dx \geq \frac{f(0)+f(1)}{2}$

B. $\int_0^1 f(x) dx \leq \frac{f(0)+f(1)}{2}$

C. $f'\left(\frac{1}{2}\right) \geq 0$

D. $f'\left(\frac{1}{2}\right) \leq 0$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

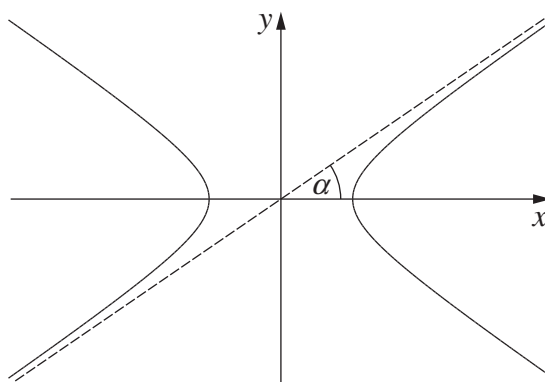
Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = 1 - \sqrt{3}i$ and $w = 1 + i$.

(i) Find the exact value of the argument of z . **1**

(ii) Find the exact value of the argument of $\frac{z}{w}$. **2**

(b) An asymptote to the hyperbola $\frac{x^2}{12} - \frac{y^2}{4} = 1$ makes an angle α with the positive x -axis, as shown. **2**



Find the value of α .

(c) Sketch the region in the Argand diagram where **2**

$$-\frac{\pi}{4} \leq \arg z \leq 0 \quad \text{and} \quad |z - 1 + i| \leq 1.$$

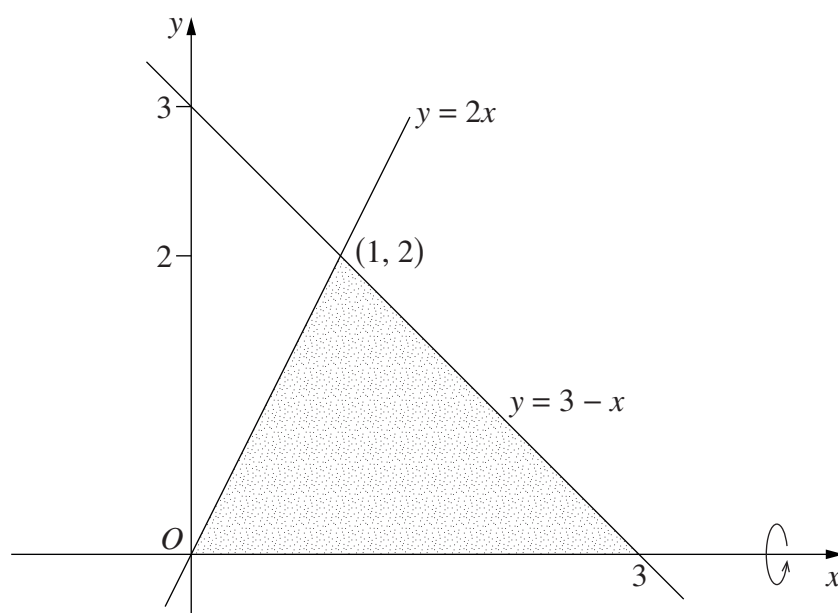
Question 11 continues on page 8

Question 11 (continued)

- (d) Using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, evaluate **3**

$$\int_0^{\frac{2\pi}{3}} \frac{1}{1 + \cos \theta} d\theta.$$

- (e) The region bounded by the lines $y = 3 - x$, $y = 2x$ and the x -axis is rotated about the x -axis. **2**



Use the method of cylindrical shells to find an integral whose value is the volume of the solid of revolution formed. Do NOT evaluate the integral.

- (f) Using the substitution $x = \sin^2 \theta$, or otherwise, evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx$. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$.

(i) Show that $f(x)$ is increasing for all x . **1**

(ii) Show that $f(x)$ is an odd function. **1**

(iii) Describe the behaviour of $f(x)$ for large positive values of x . **1**

(iv) Hence sketch the graph of $f(x) = \frac{e^x - 1}{e^x + 1}$. **1**

(v) Hence, or otherwise, sketch the graph of $y = \frac{1}{f(x)}$. **1**

(b) Solve the quadratic equation $z^2 + (2 + 3i)z + (1 + 3i) = 0$, giving your answers in the form $a + bi$, where a and b are real numbers. **3**

(c) Find $\int x \tan^{-1} x \, dx$. **3**

(d) Let $P(x)$ be a polynomial.

(i) Given that $(x - \alpha)^2$ is a factor of $P(x)$, show that **2**

$$P(\alpha) = P'(\alpha) = 0.$$

(ii) Given that the polynomial $P(x) = x^4 - 3x^3 + x^2 + 4$ has a factor $(x - \alpha)^2$, find the value of α . **2**

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Show that $\frac{r+s}{2} \geq \sqrt{rs}$ for $r \geq 0$ and $s \geq 0$. **1**

(b) Let a, b and c be real numbers. Suppose that $P(x) = x^4 + ax^3 + bx^2 + cx + 1$ has roots $\alpha, \frac{1}{\alpha}, \beta, \frac{1}{\beta}$, where $\alpha > 0$ and $\beta > 0$.

(i) Prove that $a = c$. **2**

(ii) Using the inequality in part (a), show that $b \geq 6$. **2**

(c) A particle is projected upwards from ground level with initial velocity $\frac{1}{2}\sqrt{\frac{g}{k}} \text{ m s}^{-1}$, where g is the acceleration due to gravity and k is a positive constant. The particle moves through the air with speed $v \text{ m s}^{-1}$ and experiences a resistive force. **4**

The acceleration of the particle is given by $\ddot{x} = -g - kv^2 \text{ m s}^{-2}$. Do NOT prove this.

The particle reaches a maximum height, H , before returning to the ground.

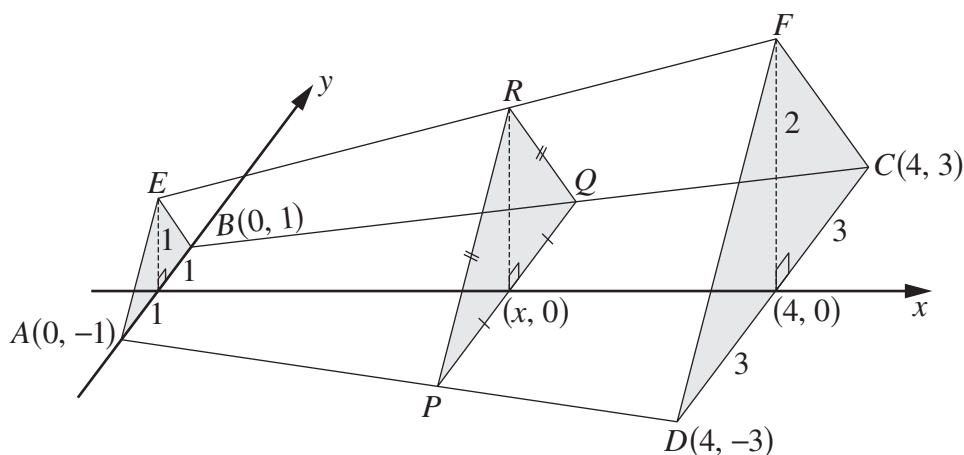
Using $\ddot{x} = v \frac{dv}{dx}$, or otherwise, show that $H = \frac{1}{2k} \log_e \left(\frac{5}{4} \right)$ metres.

Question 13 continues on page 11

Question 13 (continued)

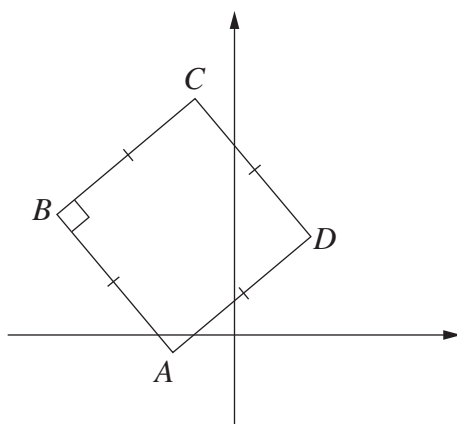
- (d) The trapezium whose vertices are $A(0, -1)$, $B(0, 1)$, $C(4, 3)$ and $D(4, -3)$ forms the base of a solid. 4

Each cross-section perpendicular to the x -axis is an isosceles triangle. The height of the isosceles triangle ABE with base AB , is 1. The height of the isosceles triangle DCF with base DC , is 2. The cross-section through the point $(x, 0)$ is the isosceles triangle PQR , where R lies on the line EF , as shown in the diagram.



Find the volume of the solid.

- (e) The points A , B , C and D on the Argand diagram represent the complex numbers a , b , c and d respectively. The points form a square as shown on the diagram. 2



By using vectors, or otherwise, show that $c = (1 + i)d - ia$.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) It is given that

$$x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2).$$

(i) Find A and B so that

1

$$\frac{16}{x^4 + 4} = \frac{A + 2x}{x^2 + 2x + 2} + \frac{B - 2x}{x^2 - 2x + 2}.$$

(ii) Hence, or otherwise, show that for any real number m ,

2

$$\int_0^m \frac{16}{x^4 + 4} dx = \ln\left(\frac{m^2 + 2m + 2}{m^2 - 2m + 2}\right) + 2 \tan^{-1}(m + 1) + 2 \tan^{-1}(m - 1).$$

(iii) Find the limiting value as $m \rightarrow \infty$ of

1

$$\int_0^m \frac{16}{x^4 + 4} dx.$$

Question 14 continues on page 13

Question 14 (continued)

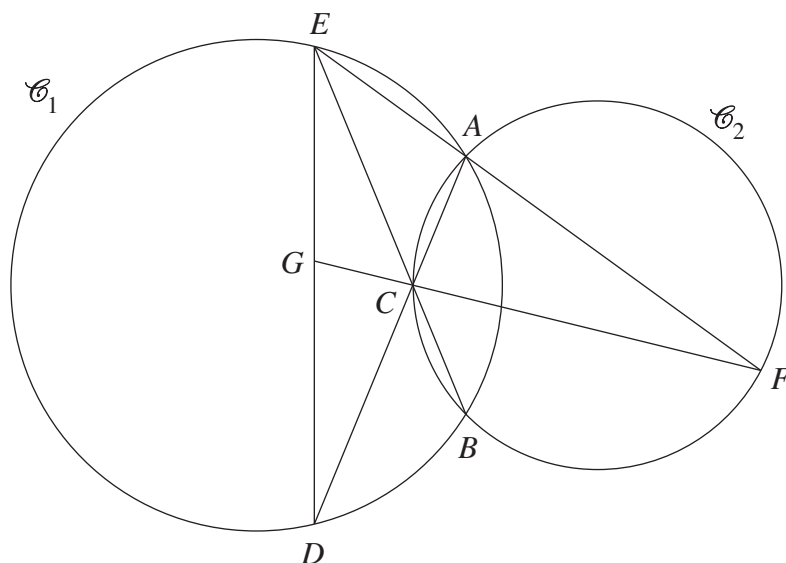
- (b) Two circles, $\mathcal{C}_1$ and $\mathcal{C}_2$, intersect at the points A and B . Point C is chosen on the arc AB of $\mathcal{C}_2$ as shown in the diagram.

The line segment AC produced meets $\mathcal{C}_1$ at D .

The line segment BC produced meets $\mathcal{C}_1$ at E .

The line segment EA produced meets $\mathcal{C}_2$ at F .

The line segment FC produced meets the line segment ED at G .



Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) State why $\angle EAD = \angle EBD$. | 1 |
| (ii) Show that $\angle EDA = \angle AFC$. | 1 |
| (iii) Hence, or otherwise, show that B, C, G and D are concyclic points. | 3 |

Question 14 continues on page 14

Question 14 (continued)

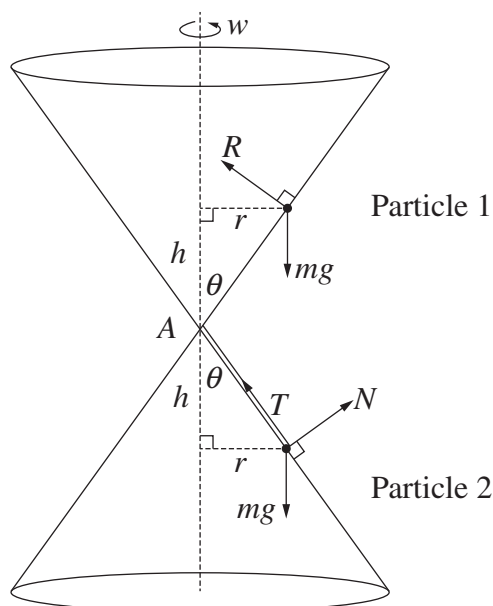
- (c) A smooth double cone with semi-vertical angle $\theta < \frac{\pi}{2}$ is rotating about its axis with constant angular velocity w .

Two particles, each of mass m , are sitting on the cone as it rotates, as shown in the diagram.

Particle 1 is inside the cone at vertical distance h above the apex, A , and moves in a horizontal circle of radius r .

Particle 2 is attached to the apex A by a light inextensible string so that it sits on the cone at vertical distance h below the apex. Particle 2 also moves in a horizontal circle of radius r .

The acceleration due to gravity is g .



- (i) The normal reaction force on Particle 1 is R . 2

By resolving R into vertical and horizontal components, or otherwise, show that $w^2 = \frac{gh}{r^2}$.

- (ii) The normal reaction force on Particle 2 is N and the tension in the string is T . 2

By considering horizontal and vertical forces, or otherwise, show that

$$N = mg \left(\sin \theta - \frac{h}{r} \cos \theta \right).$$

- (iii) Show that $\theta \geq \frac{\pi}{4}$. 2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Let $I_n = \int_0^1 x^n \sqrt{1-x^2} dx$, for $n = 0, 1, 2, \dots$.

(i) Find the value of I_1 . **1**

(ii) Using integration by parts, or otherwise, show that for $n \geq 2$ **3**

$$I_n = \left(\frac{n-1}{n+2} \right) I_{n-2}.$$

(iii) Find the value of I_5 . **1**

(b) Consider the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$, for $x \geq 0$ and $y \geq 0$, where a is a positive constant.

(i) Show that the equation of the tangent to the curve at the point $P(c, d)$ is given by $y\sqrt{c} + x\sqrt{d} = d\sqrt{c} + c\sqrt{d}$. **2**

(ii) The tangent to the curve at the point P meets the x and y axes at A and B respectively. Show that $OA + OB = a$, where O is the origin. **3**

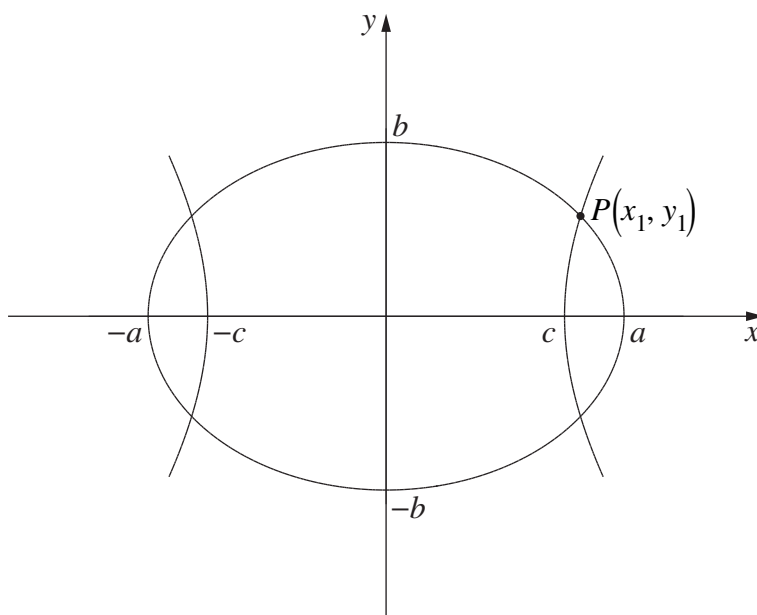
Question 15 continues on page 16

Question 15 (continued)

- (c) The ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$, has eccentricity e .

The hyperbola with equation $\frac{x^2}{c^2} - \frac{y^2}{d^2} = 1$, has eccentricity E .

The value of c is chosen so that the hyperbola and the ellipse meet at $P(x_1, y_1)$, as shown in the diagram.



- (i) Show that $\frac{x_1^2}{y_1^2} = \frac{a^2 c^2}{(a^2 - c^2)} \times \frac{(b^2 + d^2)}{b^2 d^2}$. 2
- (ii) If the two conics have the same foci, show that their tangents at P are perpendicular. 3

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Let $\alpha = \cos \theta + i \sin \theta$, where $0 < \theta < 2\pi$.

(i) Show that $\alpha^k + \alpha^{-k} = 2 \cos k\theta$, for any integer k . 1

Let $C = \alpha^{-n} + \cdots + \alpha^{-1} + 1 + \alpha + \cdots + \alpha^n$, where n is a positive integer.

(ii) By summing the series, prove that $C = \frac{\alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})}{(1 - \alpha)(1 - \bar{\alpha})}$. 3

(iii) Deduce, from parts (i) and (ii), that 2

$$1 + 2(\cos \theta + \cos 2\theta + \cdots + \cos n\theta) = \frac{\cos n\theta - \cos(n+1)\theta}{1 - \cos \theta}.$$

(iv) Show that $\cos \frac{\pi}{n} + \cos \frac{2\pi}{n} + \cdots + \cos \frac{n\pi}{n}$ is independent of n . 1

(b) The hyperbola with equation 2

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

has eccentricity 2.

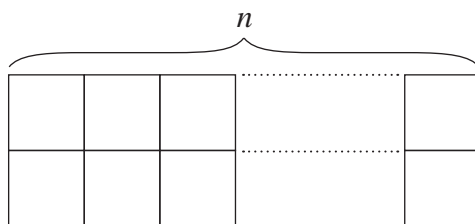
The distance from one of the foci to one of the vertices is 1.

What are the possible values of a ?

Question 16 continues on page 18

Question 16 (continued)

- (c) A 2 by n grid is made up of two rows of n square tiles, as shown.



The tiles of the 2 by n grid are to be painted so that tiles sharing an edge are painted using different colours. There are x different colours available, where $x \geq 2$.

It is NOT necessary to use all the colours.

Consider the case of the 2 by 2 grid with tiles labelled A , B , C and D , as shown.

A	C
B	D

There are $x(x - 1)$ ways to choose colours for the first column containing tiles A and B . Do NOT prove this.

- (i) Assume the colours for tiles A and B have been chosen. There are two cases to consider when choosing colours for the second column. Either tile C is the same colour as tile B , or tile C is a different colour from tile B . 2

By considering these two cases, show that the number of ways of choosing colours for the second column is

$$x^2 - 3x + 3.$$

- (ii) Prove by mathematical induction that the number of ways in which the 2 by n grid can be painted is $x(x - 1)(x^2 - 3x + 3)^{n-1}$, for $n \geq 1$. 2
- (iii) In how many ways can a 2 by 5 grid be painted if 3 colours are available and each colour must now be used at least once? 2

End of paper

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REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

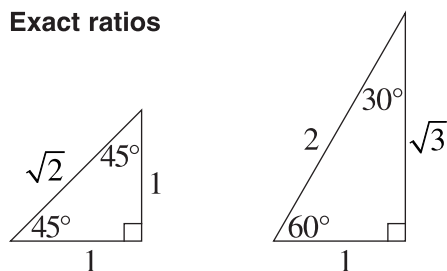
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$